

Oligopoly

2nd

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Oligopoly: It is an industry that contains only a few competing firms. Each firm has enough market power to prevent it being price taker. Oligopolists are aware of the interdependence among the decisions made by the various firms in the industry and they engage in rivalrous and strategic behaviour. Examples are oil, automobile, steel, airline.

Interdependence, because there are only few firms under oligopoly, each firm will have to take account of others. This means that they are mutually dependent. Each firm is affected by its rival's actions. If the firm changes the price or the amount of its advertising, the sale of its rival will be affected. The rivals may then respond by changing their price or advertising. No firm can afford to ignore the actions and reactions of other firms in the industry.

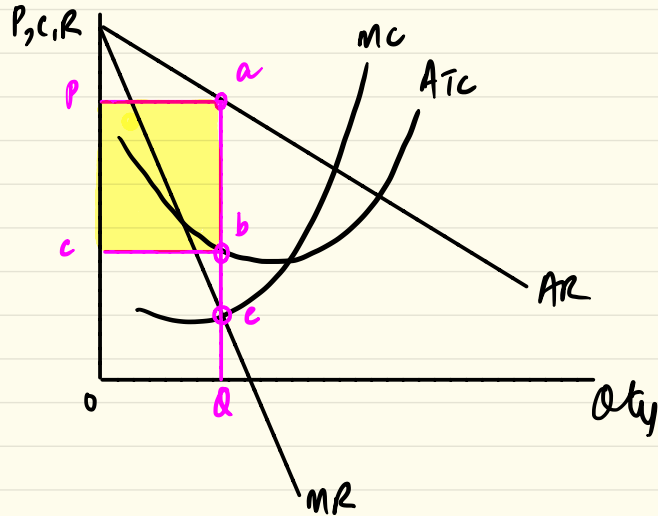
In economics, a concentration ratio is a measure of the total output produced in an industry by a given number of firms in the industry. Concentration ratios are usually used to show the extent of market control of the largest firms in the industry and to illustrate the degree to which an industry is oligopolistic. For e.g. four firms concentration ratio is 25% which means that the largest four firm's market share totals 25%.

Oligopoly

- * Few competing firms
- * 4 to 5 large firms
- * Interdependence
- * Rivalrous and strategic
- * Standardized and differentiated products (Diversity)
- * E.g. Oil, automobile, steel, airline.
- * Economies of scale
- * Merger and takeovers
- * Capital required
- * Price rigidity (Kink)
- * Collusion (cartel)

Collusive Oligopoly:-

- * When firms agree to cooperate in order to restrict output and raise prices
- * Collusive behaviour may occur with or without an actual agreement to collude



Cartels: Cartels have already been referred to in the context of a collusive oligopoly market structure when it was pointed out that there could be a degree of collusion between firms in an oligopoly market, operating together through a cartel.

A cartel is a grouping of producers that work together to defend their shared interests. Cartels are created when a few large producers decide to collude, for e.g. to fix prices for members so that competition on price is avoided. They can also restrict output released onto the market. An e.g. of such cartel is OPEC, the organization of the petroleum exporting countries. It was established in 1960 and consists of 13 countries. It controls about 50% of global oil production and about 80% of the world's oil reserves.

Conditions for effective cartel:

- * **Barriers to entry**: a cartel is more likely to be effective when there are high barriers to entry into a market or an industry.
- * **Control over both price and output**: a cartel is also more likely to be effective if it can not only fix prices for members, but also restrict output.
- * **Setting rules**: a cartel will need to set rules governing the behaviour of members; if members follow these rules, risks that would exist without a cartel are reduced.
- * **Policing rules**: a cartel is also more likely to be effective when all members can be 'policed' in some way to ensure that the rules are being obeyed.

Consequences of cartel

- * **Production of shared interest**: members of cartel work together to defend their shared interest
- * **Avoidance of price competition**: cartels fix prices for their members, so that competition on the basis of price is avoided.
- * **Effect on revenue**: where a cartel is able to control both price and output, revenue will be substantially increased
- * **Control of market**: the existence of a cartel will give members a dominant position in a market; for example, OPEC controls 80% of the oil producing market

Negative consequences of a cartel for consumers

- High prices**: the members of a cartel can all raise prices together, which reduces the price elasticity of demand for any particular member and makes products more expensive for consumers.
- Restricted output**: members of a cartel may agree to limit output onto a market, such as through a quota system, reducing competition.
- Lack of transparency**: members of a cartel may agree to hide prices or withhold information, such as in relation to hidden charges to consumers involved in certain transactions.
- Carving up a market**: members of a cartel may collectively agree to break up a market into regions and not compete in each other's area, reducing choice for consumers.

Non-collusive Oligopoly (The kink demand curve): It is an operational tool for the determination of the equilibrium in the oligopolistic markets. The kink demand curve is made out of the firm's own demand curve and the market demand curve. The firm's own demand curve is elastic because of availability of substitutes and industry's demand curve is inelastic because there is no close substitute for the industry's product. The upper section has higher elasticity because it is part of firm's own demand curve and lower part has low elasticity because it is part of industry's demand curve. Due to the kink in the demand curve, marginal revenue is discontinuous at the level of output corresponding to the kink.

Diagram #1

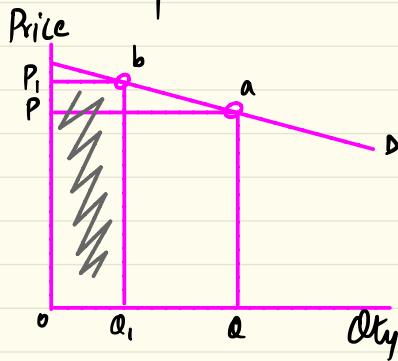


Diagram #2

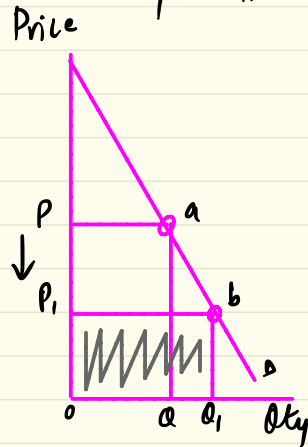


Diagram #3

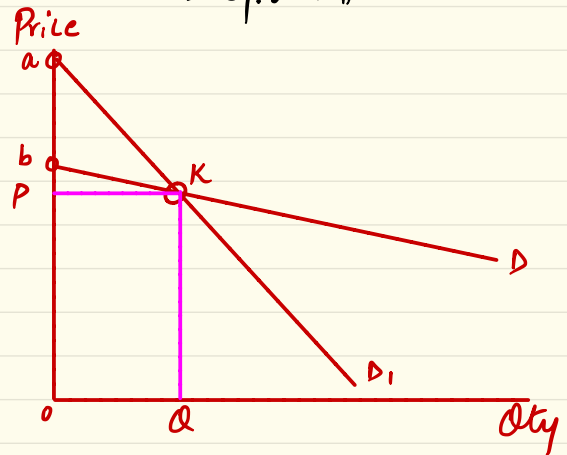


Diagram #1 :- This explains the concept of elastic demand, where responsiveness is high because of availability of substitutes. If there is an increase in price, it will contract the quantity demanded by greater percentage, this reduces total revenue. This will not be an effective policy.

Diagram #2 :- This explains the concept of inelastic demand, where responsiveness is low because of less availability of substitutes. If there is an increase in price, quantity will not extend by greater percentage. For almost same quantity, consumers will be required to pay less, which benefits them as it reduces total expenditure of consumers, which is a disadvantage for producer as they see reduction in total revenue. Again not an effective policy.

Diagram #3 :- This diagram is a combination of diagram 1 and diagram #2. In an industry, a firm's demand curve is elastic as there is a lot of firms in the industry. While talking about industry's demand curve which comprises largest firms of that industry, their demand curve will be inelastic, as there is no such alternative of the industry.

In the diagram 'bK' is firm's demand curve $\rightarrow$ Elastic demand
Industry demand curve is 'KD₁' $\rightarrow$ Inelastic demand

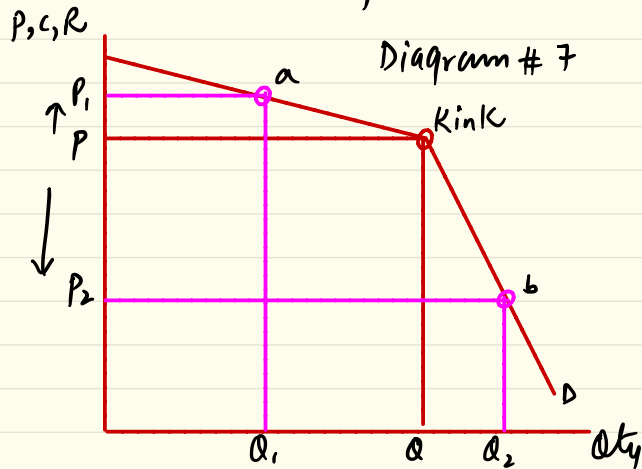
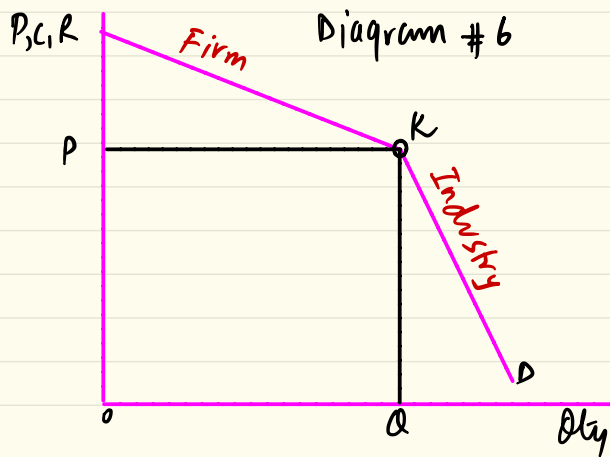
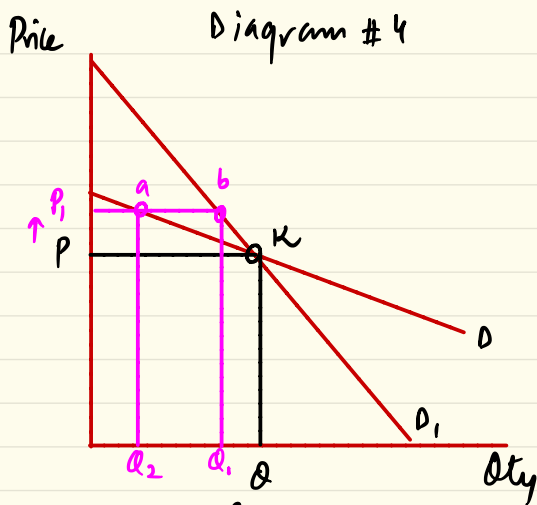
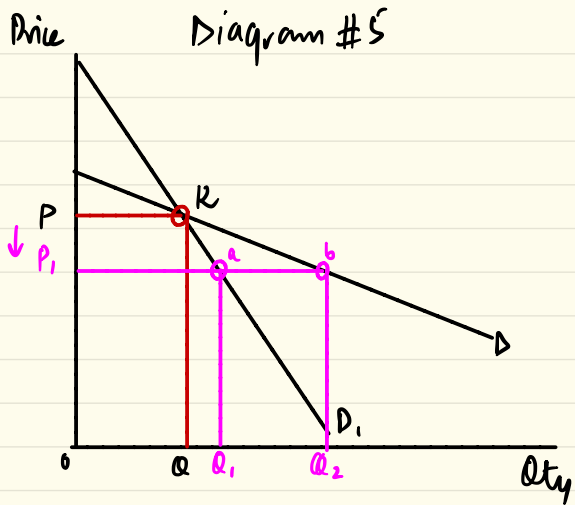


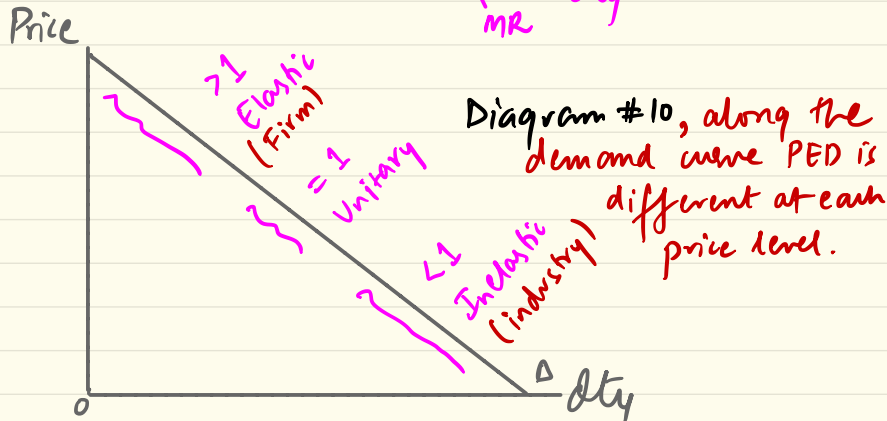
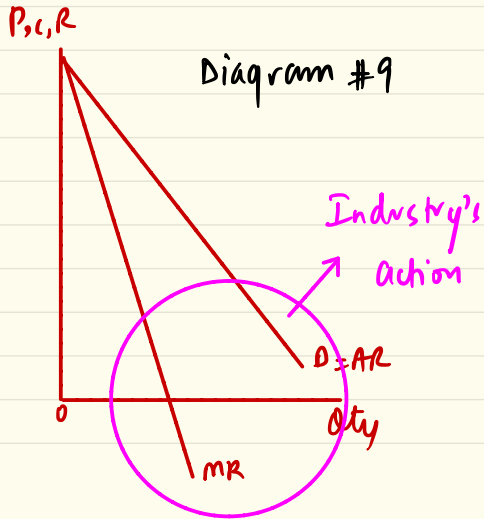
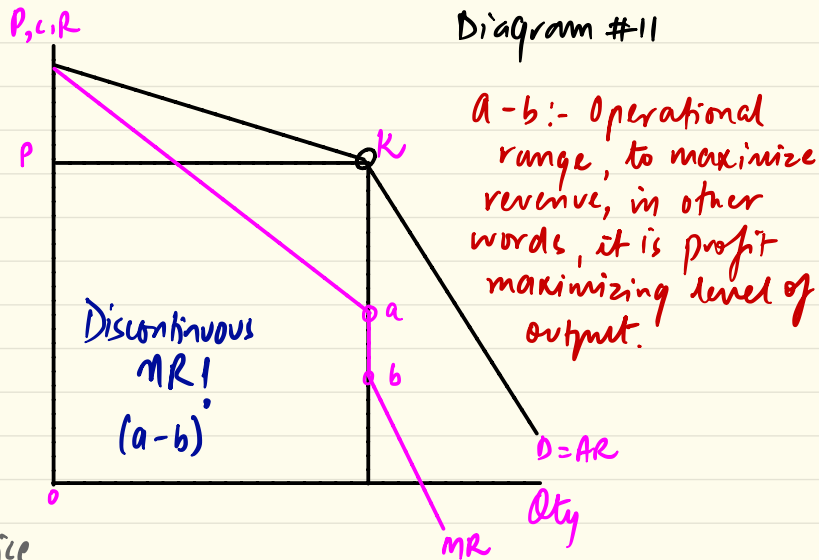
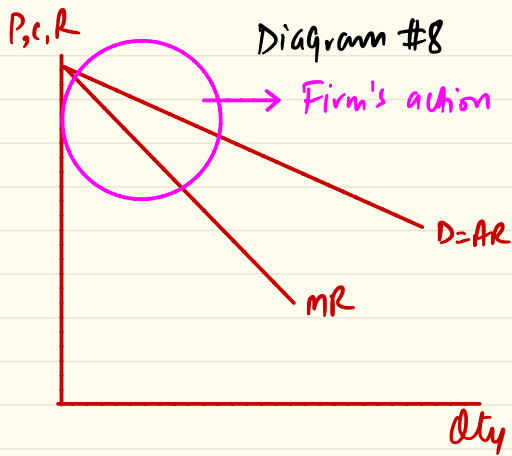
Diagram #4:- In an oligopoly market structure, with few large firms, if one firm decides to increase the price, thinking it will increase total revenue, but other oligopolist will not respond to this situation. Thus action of one firm will be considered as elastic demand, rather than being inelastic. This is why an increase in price by smaller percentage will contract quantity demanded by greater percentage, which reduces total revenue.

Diagram #5:- While reduction in price by a firm, thinking that other firms will not follow, and having an expectation of increased revenue. But this time, other firms will follow and reduce the price, if every firm reduces the price and bring it equal, consumers will not switch and this will reduce their revenue as quantity demanded will not exceed by greater percentage, this reduces revenue.

One firm's action not followed by others — Elastic demand ($\text{Price} \uparrow \Rightarrow Q \downarrow \text{TR} \downarrow$)

Industry's action — Inelastic demand ($\text{Price} \downarrow \Rightarrow Q \uparrow \text{TR} \downarrow$)

Diagram #6 #7:- This is how kink demand curve (Price rigidity) is established. Upper part of the demand curve, elastic demand (firm's action), lower part of the demand curve, inelastic demand (industry's action). If prices are increased or decreased, total revenue reduced, thus prices remain unchanged at 'Kink'.



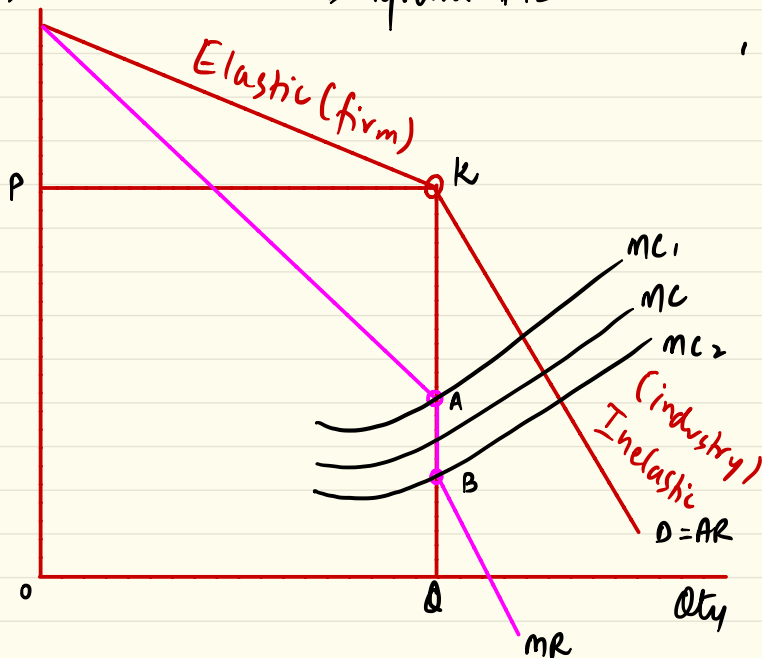
P, C, R

Diagram #12

'Kink demand curve'

'price rigidity'

'Non-collusive Behaviour'



Profit maximizing level of output, where marginal cost = marginal revenue between the range of 'A-B'.

Game theory: It involves the analysis of the strategies adopted by rational players (individuals, groups, firms or even nations) in situations when each player must make decisions in the knowledge that the final outcome (pay off) will depend on the reactions of other players and where the first player knows the other is thinking in the same way.

All games, however, have the common features that they are played out in an environment of uncertainty, interdependence and often conflicts of interest amongst the players

Here we concentrate on relatively straightforward two-player zero-sum games where one player's gain must equal the other player's loss.

Prisoner's dilemma in Oligopolistic markets and the two player matrix

The most famous and often quoted e.g. of game theory strategy is the prisoner's dilemma. In this example, two individuals, Adam and Ben, have been arrested on suspicion of having committed a very serious crime, which carries a maximum prison sentence of ten years.

Adam and Ben are being detained in separate parts of the prison and have no way of communicating with one another.

The chief of police meets with each in turn and speaks to them in the same way:

"I have sufficient evidence to send both of you to prison for three years if neither of you confesses. However, if you alone are prepared to confess to the crime you will receive a light sentence of one year and your partner will receive a prison sentence of ten years.

If both of you confess, you will both receive a sentence of 5 years"

What should Adam and Ben do? What is the best strategy for each in these circumstances?

		Adam	
		Not confess	Confess
Ben	Not confess	A Adam 3 yrs Ben 3 yrs	B Adam 1 yr Ben 10 yrs
	Confess	C Adam 10 yrs Ben 1 yr	D Adam 5 yrs Ben 5 yrs

Let us begin with Adam.

It is best that I confess because in the best case scenario I will receive a light sentence of one year if Ben doesn't confess and in the worst case scenario I will receive a sentence of 5 years rather than 10 years in prison if Ben also confesses.

Ben will follow the same path of reasoning and also conclude that the best strategy is to confess.

The final result is that both Adam and Ben will confess and receive a sentence of 5 years each as illustrated in quadrant D of the matrix.

Clearly this is not the best outcome for Adam and Ben. The best outcome would be for them to co-operate or collude and both agree to not confess as illustrated in quadrant A, which would give them a lighter sentence of three years. However, in the absence of any ability to communicate with one another they would each have to trust in the other's willingness to stick to the deal.

Other pricing policies

Limit pricing refers to a situation in which price is below the profit maximizing price. For e.g. a firm may decide to limit price in an attempt to discourage new firms from entering an industry and so protecting the position of the firm.

Oligopoly can sometimes give rise to predatory pricing. This is where a firm charges a price that is lower than those of competitors in a deliberate attempt to force other firms out of the industry.

Price leadership is most likely to exist in an Oligopoly market structure where there is some degree of collusion between the firms in the market. In this situation, firms in the market will follow the price leadership of one firm. The objective is to maximize the profits of all the firms by behaving as if they were one monopolistic firm. This agreement on price could be informal and formal.

Question

Q1 Write a summary on Oligopoly? 25 marks

Use notes booklet

Ans:- * Define Oligopoly + feature

paragraph given

* Non-collusive Oligopoly, Kink demand curve (Diagram)

* Game theory, pay off matrix, Prisoner's dilemma

* Collusive Oligopoly, Cartel (Diagram)

* Price leadership

Use the paragraph

Draw the matrix and explain in detail

paragraph given